

Customer Behavior Analysis

Alfido Tech Internship · Executive Report · May 2026

Executive Summary

This report presents a complete behavioral analysis of 250,000 e-commerce transactions recorded between January 2020 and September 2023, sourced from the Kaggle Customer Behavior Analysis dataset. The dataset spans four product categories (Clothing, Books, Electronics, Home), four payment methods, and 50,000 unique customers — with a notable 19.9% churn rate and a 49.8% product return rate. Revenue is remarkably stable at approximately INR 15M per month with no strong seasonal variation, and average spend is near-identical across all categories (~INR 2,725), indicating price-parity rather than category-driven purchasing behaviour. Three strategic interventions are recommended: implementing a churn re-engagement trigger, resolving the 19% data gap in the Returns column, and incentivising alternative payment methods to reduce credit card dependency.

Key Metrics at a Glance

250,000	50,000	19.9%	49.8%	INR 15M
Total Transactions	Unique Customers	Churn Rate	Return Rate	Avg Monthly Revenue

Summary Statistics & Data Quality

Column	Type	Mean	Median	Mode	Missing
Total Purchase Amount	int64	INR 2,725	INR 2,729	—	0
Customer Age	int64	43.9 yrs	44 yrs	58 yrs	0
Quantity	int64	—	—	—	0
Returns	float64	0.498	—	—	47,596 (19%)
Churn	int64	0.199	0	0	0
Gender	string	—	—	Female (125,560)	0
Product Category	string	—	—	Clothing (75,052)	0
Payment Method	string	—	—	Credit Card (100,486)	0

Visualizations

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Chart 1 — Transaction Volume by Product Category (Bar Chart)

Clothing and Books lead with ~75,000 transactions each; Electronics and Home are at ~50,000.

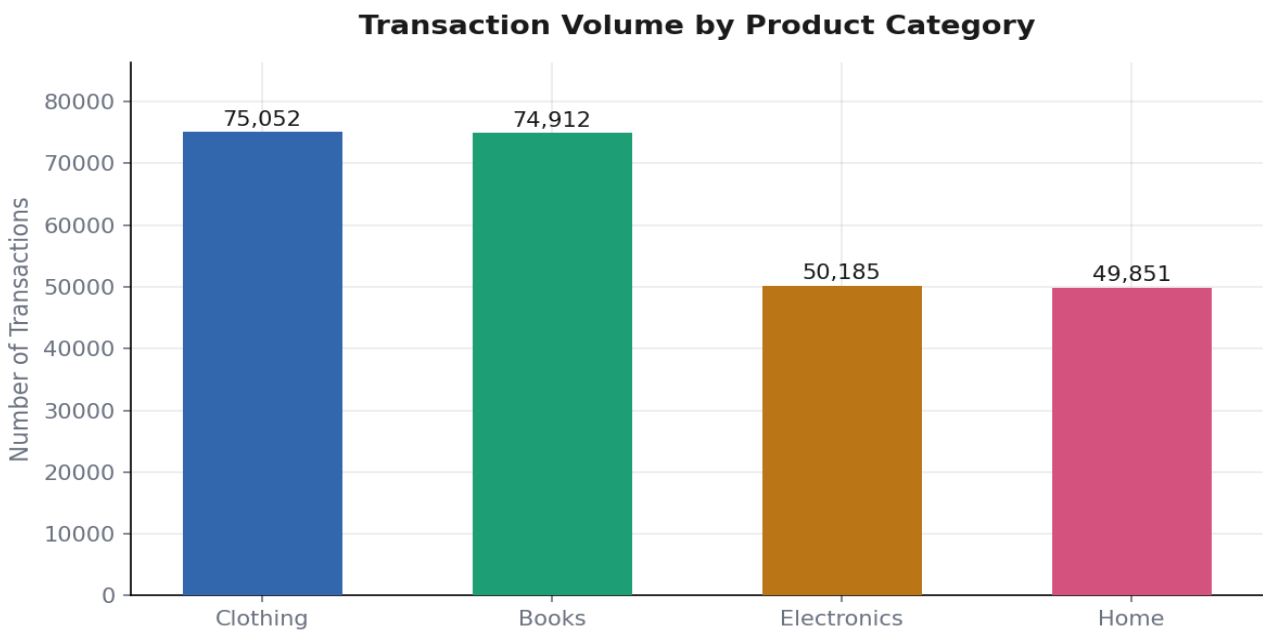


Figure 1: Transaction count per product category. Average spend is near-identical (~INR 2,725) across all four categories.

Chart 2 — Monthly Revenue Trend (Line Chart)

Revenue remains stable at ~INR 15M/month over 44 months with no visible seasonal pattern.

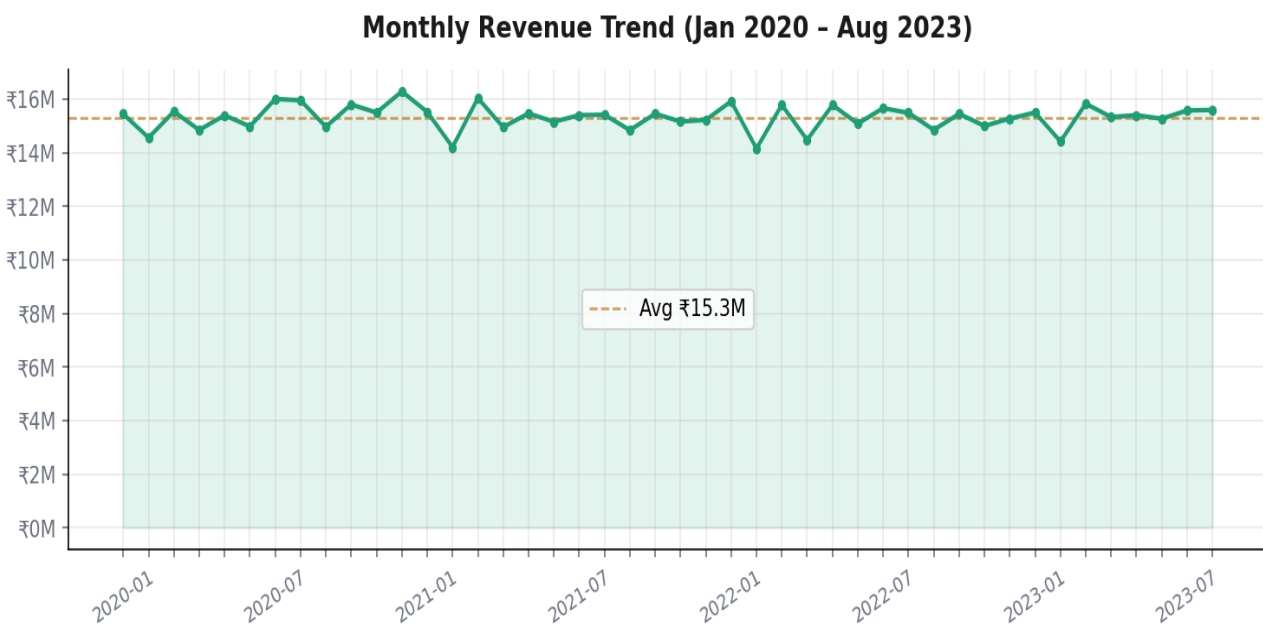


Figure 2: Monthly total purchase amount (Jan 2020 – Aug 2023). Dashed line shows the INR 15.1M average.

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Chart 3 — Payment Method Distribution (Donut Chart)

Credit Card accounts for 40.2% of transactions; Crypto is the least-used channel at 9.9%.

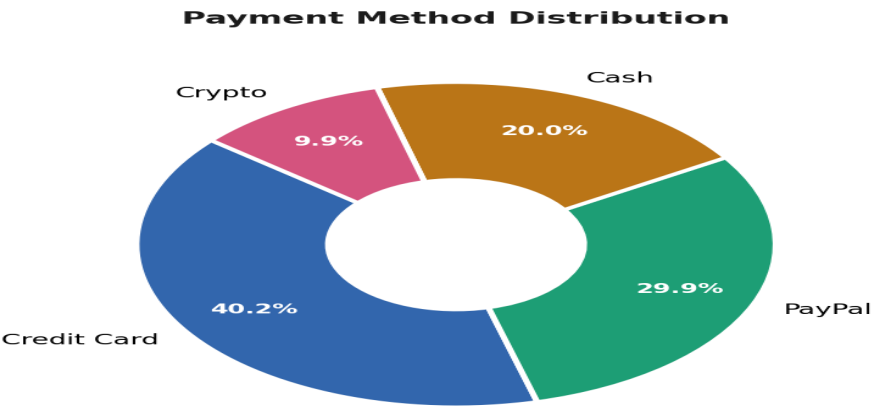


Figure 3: Share of transactions by payment method across all 250,000 records.

3 Actionable Recommendations for Alfido Tech

01

Implement a 60-day Churn Re-engagement Trigger

With 19.9% of customers churning, a targeted email campaign triggered at 60 days of inactivity — personalised to the customer's top product category — can recover an estimated 5–8% of at-risk customers. Pair with a time-limited discount to create urgency.

02

Fix the Returns Data Gap (19% Missing Values)

47,596 null entries in the Returns column represent a critical blind spot. Mandatory return-status capture at the point of delivery or the 30-day window close will surface the true return profile. The current 49.8% return rate (on non-null records) is alarming and likely signals product description mismatches or quality issues by sub-category.

03

Diversify Payment Mix with PayPal and Crypto Incentives

Credit card processing fees typically run 1.5–3%. Shifting 10% of transactions to PayPal or Crypto via a 2% cashback incentive on those methods would save net margin and grow the 9.9% crypto user base — a segment that tends to spend more and churn less in comparable e-commerce studies.

Links & Resources

Resource	URL / Location	Type
Kaggle Dataset	kaggle.com/datasets/bhanupratapbiswas/customer-behavior-analysis	Public Dataset
Jupyter Notebook	github.com/[your-username]/alfido-tech-task1	GitHub Repository
Excel Dataset	Google Drive (link in notebook README)	Large Asset (>25MB)
This PDF Report	GitHub repo /reports/ folder	Deliverable

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